



**CALICUT UNIVERSITY – FOUR-YEAR UNDER
GRADUATE PROGRAMME (CU-FYUGP)**

BSc PHYSICS HONOURS

Programme	B.Sc. Physics Honours				
Course Title	PRINCIPLES OF RESEARCH METHODOLOGY				
Type of Course	Core in Major				
Semester	VIII				
Academic Level	400 - 499				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-	-	60
Pre-requisites	Major courses in first 6 semester				
Course Summary	This course equips students with the critical thinking skills and scientific methods to distinguish facts, design experiments, and analyze research.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Distinguish between scientific facts, generalizations, and pseudo-science, understanding the social nature of scientific	U	C	Instructor-created exams / Quiz

	activity and its role in democratic development.			
CO2	Critically evaluate the limitations of science, including its underlying assumptions and challenges in defining reality and rationality.	E	P	Instructor-created exams / Quiz
CO3	Explain the key concepts of description, causality, prediction, and explanation in science, along with the role of mathematics in scientific endeavors.	U	C	Instructor-created exams / Quiz/Viva
CO4	Differentiate between hypotheses, theories, and laws, critically evaluating the processes of verification, falsification, acceptance, and peer review in the scientific method.	An	P	Instructor-created exams / Home Assignments
CO5	Apply principles of measurement, including operationalization (variables and indicators), to scientific research. Students will be able to evaluate the validity, reliability, and reproducibility/replicability of measurements and identify potential sources of error.	Ap	P	Home Assignments
CO6	Design and analyze experiments, understanding the roles and limitations of experimentation, including natural, manipulative, and comparative approaches. Students will be able to assess the validity and reliability of experiments using appropriate epistemological strategies.	C	M	Seminar/Viva
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (48 +12)	Marks (70)
I	METHODOLOGY OF SCIENCE		15	25
	1	Science as facts, science as generalization, Some distinctions when describing science	3	
	2	Science as a social activity, scientific revolutions and paradigms	2	
	3	Science and pseudo-science	1	
	4	Science and democratic development		
	5	The limitations of science-presuppositions, fundamental questions on reality, Rationality	2	
	6	Description, Causality, Prediction and Explanation in science -	2	
	7	Mathematics and science	1	
	8	Hypothesis, Theories and laws	2	
	9	Verification, Falsification, Acceptance, Peer Review in Science - Scientific method	2	
	Sections 2.2.1 – 2.2.5, 2.3.1, 2.4.1, 2.5.1 – 2.5.4, 2.6.1 – 2.6.4, 2.8.1 – 2.8.4, 3.1 – 3.3, 4.1 – 4.4, 7.1 of Book 1			
II	MEASUREMENT		9	15

	10	Processes, Instruments and Operationalization, Operationalization (Variables and Indicators),	3	
	11	Criteria in Measurement, Validity, Reliability, Reproducibility/Replicability	3	
	12	Measurement Error, Potential Sources of Measurement Error, Random and Systematic Errors	3	
	Sections 5.2.1 – 5.2.2, 5.2.3 of Book 1			
III	EXPERIMENTATION		12	15
	13	The Roles and Limitations of Experimentation	2	
	14	Natural Experiments, Manipulative Experiments, Comparative Experiments	3	
	15	Experimentation and Research, Conducting Experiments	2	
	16	Validity and Reliability in Experimentation, Epistemological Strategies	3	
	15	Design of Experiments	2	
	Sections 6.1.1 – 6.1.2, 6.1.3, 6.2, 6.3, 6.4 of Book1			
IV	SCIENTIFIC METHOD AND DESIGN OF RESEARCH		12	15
	17	The Scientific Method, Research Design, Components,	2	
	18	Research Design and Proposal, Purpose of Proposal, Proposal Structure	3	
	19	Conceptual Framework (or Literature Review)	2	
	20	Research Questions/Hypotheses	1	
	21	Methods/Methodology	2	

	22	Validity, Concluding sections to proposal	2	
	Sections 7.1 – 7.2, 7.2.1, 7.2.2 of Book 1			
V	OPEN ENDED MODULE: RESEARCH		12	
		Basic, Applied and Evaluation Research, Multidisciplinary and Interdisciplinary Research, The Value of Having Research Skills, Formulating a Research Problem, Research in Relation to Teaching and Publishing. Ethics and Responsibility in Scientific Research, Ethics, Western and Eastern Perspectives on the Source of Ethics, Unethics, Guidelines for Ethical Practices in Research, Plagiarism, Integrity of data, Use and misuse of data, Ownership of and access to data, Obligation to report, Conflict of Interest, From Unethics to Ethics in Research, The Responsibility of Scientists and of Science as an Institution		
Books and References: The Aims, Practices and Ethics of Science by Peter Pruzan; Springer International Publishing Limited (Book 1)				

Mapping of COs with PSOs and POs :

	PSO 1	PSO 2	PSO 3	PSO 4	PSO 5	PSO 6	PO1	PO2	PO3	PO4	PO 5	PO 6	PO 7
CO 1	0	0	0	0	1	0	2	2	0	0	0	2	1
CO 2	0	0	0	0	1	0	2	2	0	0	0	2	1
CO 3	1	0	1	0	1	1	2	2	0	0	1	2	1
CO 4	1	0	1	0	1	1	2	2	0	0	1	2	1
CO 5	0	1	0	0	0	0	2	2	1	1	1	2	1
CO 6	0	1	0	0	0	0	2	2	0	0	0	2	1

Correlation Levels:

Level	Correlation
0	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Discussion / Seminar
- Internal Theory / Practical Exam
- Assignments / Viva
- End Semester Exam (70%)

Mapping of COs to Assessment Rubrics

	Internal Theory / Practical Exam	Assignment / Viva	Practical Skill Evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓	✓		✓
CO 3	✓	✓		✓
CO 4	✓	✓		✓
CO 5	✓	✓		✓
CO 6	✓	✓		✓

MAJOR ELECTIVE COURSES